

Muhammad Rafi Athallah

Electronic Engineering Student

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PROFILE

Electronic Engineering student at Semarang State Polytechnic Department of Electrical Engineering, Electronics Engineering Technology program. Focused on embedded systems, microcontrollers, robotics, Internet of Things (IoT), computer vision, and web development. Actively building hardware and software solutions through academic projects and student organizations.

EDUCATION

POLITEKNIK NEGERI SEMARANG (POLINES)

2023 – Present

Bachelor of Applied Engineering in Electrical Engineering – Electronics Engineering Technology

- Focus on embedded systems, industrial automation, IoT, robotics, control systems, and electronic system design
- Admitted through independent (mandiri) pathway

SMA NEGERI 1 TEGAL

2020 – 2023

Natural Sciences (IPA)

- Focused on Mathematics, Physics, Chemistry, and Biology

EXPERIENCE

Electronic Team Member | Polines Robotics Evolution

2025 – Present

- Designed PCBs, assembled electronic circuits, integrated sensors and microcontrollers, and performed hardware troubleshooting for robotic systems
- Collaborated with mechanical and programming teams to develop reliable electronic solutions
- Contributed to the team's qualification for the 2025 Indonesian Robot ABU Indonesia (KRAI) Competition

Program Director Member | New PLBS FM

2023 – 2025

- Provided technical support for broadcasting operations including computer system maintenance, network connectivity, and audio equipment management
- Assisted in managing broadcasting activities and ensured smooth operation of digital media infrastructure during radio programs and events

SELECTED PROJECTS

IoT Water Quality Monitoring System

2025

Role: Electrical & Web Developer | Partner: Academic Project

- IoT-based water quality monitoring system measuring pH, turbidity, and temperature in real time, integrated with a web dashboard for live data visualization and threshold configuration.
- <https://instrument-project-6bafa.web.app>

DOTclock: IoT Digital Clock

2025

Role: Project Lead | Partner: PMW

- IoT-based digital clock with automatic internet time synchronization and LED dot-matrix display. Covered hardware design, embedded programming, and full system integration.

PID-Based Water Level Control System

2025

Role: Electrical & Hardware Programmer | Partner: Academic Project

- Developed the electrical system and embedded control program for a PID-based water level control system, including hardware integration, sensor and water pump interfacing, and real-time data acquisition.

BrightLink: Smart Home System

2025

Role: Full-Stack IoT Developer | Partner: Personal Project

- Personal IoT project for smart lighting control and real-time temperature and humidity monitoring, covering electrical system, embedded firmware, and web application development.

UKM E-Voting System

2025

Role: Web Developer | Partner: New PLBS FM

- Web-based e-voting application supporting transparent student organization chairman elections, featuring secure authentication, candidate management, online voting, and real-time vote counting.
- <https://pemilihan-gm.web.app>

Soma Kopi POS & Ordering System

2025

Role: Web Developer | Partner: Soma-Kopi

- Responsive web-based point-of-sale and ordering system for a coffee shop, including an interactive dashboard, digital menu, and ordering workflow for customers and administrators.
- <https://soma-kopi.web.app>

PLBS FM Profile Website

2024

Role: Web Developer | Partner: New PLBS FM

- Responsive website introducing the student organization, presenting activity information, official social media channels, and media partner requirements.
- <https://newplbsfm.github.io/PLBS-FM/>

SKILLS & ABILITIES

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|---|--|
| · Embedded Systems & Microcontrollers (Raspberry Pi, Arduino, Pico) | · Web Development (Firebase, HTML/CSS/JS) |
| · Hardware Design & PCB Design | · Project Management & technical documentation |
| · IoT – sensor integration, dashboard, real-time monitoring | · Electronic system analysis |
| | · English (active) |

ORGANIZATIONS & ASSOCIATIONS

- Electronic Team – Polines Robotics Evolution (2025 – Present)
- Program Director Member – New PLBS FM (2023 – 2025)